

Malware Types & Capabilities

The Malware Family Tree

Type	How It Spreads/Holds	Key Trait
Virus	Infects files/self-reproducing	Needs user action to replicate
Worm	Self-propagates over network	Spreads without user help
Trojan	Disguised as useful	Needs user to run it
RAT	Remote backdoor	Full remote control
Ransomware	Encrypts + demands payment	Business-destroying
Spyware/Adware	Harvests data/bombards ads	Stealth monetization
Rootkit	Hides itself deep	Evades detection (kernel/BIOS)
Bootkit	Infects boot chain	Survives reboot, pre-OS
Keylogger	Records keystrokes	Credential theft
Wiper	Destroys data	Sabotage (e.g., WhisperGate)

Classic Examples (Learn the Stories)

- ILOVEYOU (2000) – macro virus sent by email; ~\$10B damages
- Conficker (2008) – worm exploiting unpatched SMB; millions infected
- Stuxnet (2010) – state-created worm targeting Iranian uranium centrifuges; sabotage
- WannaCry (2017) – ransomware + EternalBlue (leaked NSA exploit); global outage, NHS crippled
- NotPetya (2017) – disguised as ransomware, actually a wiper; ~\$10B global damage

SolarWinds (2020) – supply-chain backdoor (trojanized update)

Malware Capabilities (What It Can Do)

Manipulation of files/data – encrypt, delete, modify

Command & Control (C2) – phone home for instructions

Spying – record keys, screen, camera, mic

Lateral movement – spread internally

Persistence – reinstall across reboots (registry, cron, services)

Evasion – anti-VM, anti-sandbox, code obfuscation (packers, polymorphism)

The Payload Chain

Malware rarely acts alone:

Delivery – phishing, web, USB, exploit

Execution – macro, LNK, loader

Persistence – autostart

C2 – beacon out

Do the job – steal, encrypt, spy

Blue-team lens: each of those steps is a detection point. You don't need to catch the payload – a suspicious beacon at step 4 is equally actionable.

Ransomware in Depth (2020s)

Encryption of documents (AES fast encrypt, RSA for keys)

Double/triple extortion – ALSO exfiltrating + threatening publication to pressure payment

RaaS economy – affiliates rent infrastructure from gangs (LockBit, BlackCat etc.)

The defense: airtight, off-line, tested backups + EDR + user hardening + fast IR

The most important malware fact: you will never "see" most malware in action; you will see its *effects* (a file, a process, a network beacon,

a ransom note). Analysis is all about connecting clues to the family and the playbook.

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